

AST 6007-01 Spacecraft Dynamics and Control

Homework #2

Due Oct 6 2021, 10:00AM

Consider the three dimensional motion of a spacecraft near the Earth-Moon Lagrangian librations point L_2 , based on the circular restricted three-body model. Then spacecraft orbit is determined by the gravity of the Earth, the gravity of the Moon, and by the propulsive forces that act on the spacecraft, expressed in the rotating barycentric coordinate frame.

1. What are the nonlinear differential equations, expressed in the barycentric coordinates, that describe the dynamics of the spacecraft under the influence of the gravity of Earth, the gravity of the Moon, and propulsive forces? Specify all the numerical constants in the problem.
2. If there is no propulsive force on the spacecraft, show that the spacecraft is in equilibrium with respect to the barycentric coordinate frame at the liberation point L_2 . What are the barycentric coordinates of this liberation point? Describe the corresponding motion of the spacecraft with respect to an inertial frame.
3. Consider perturbations of the spacecraft from the liberation point L_2 . What are the linear differential equations that describe the in-plane and out-of-plane perturbation dynamics of the spacecraft under the influence of the gravity of Earth, the gravity of the Moon, and propulsive forces? What are the eigenvalues of the in-plane and out-of-plane linear dynamics?
4. Suppose that the spacecraft has initial position and velocity with respect to the rotating coordinate frame with origin at L_2 given by

$$\begin{aligned}\Delta x &= 0km, & \Delta \dot{x} &= -0.005966km/sec, \\ \Delta y &= -3,505km, & \Delta \dot{y} &= 0km/sec \\ \Delta z &= 0km, & \Delta \dot{z} &= 0km/sec\end{aligned}$$

Suppose there are no propulsive forces acting on the spacecraft. Based on the linear perturbation dynamics in Problem 3, determine the position vector and the velocity vector of the spacecraft in the rotating coordinate frame with the origin at L_2 . Plot Δy vs. Δx during one orbital revolution of the Earth-Moon system; in this motion a Halo orbit? Describe the motion of the spacecraft with respect to an inertial frame.

5. Using the linear differential equations in Problem 3, are the spacecraft in-plane dynamics controllable using propulsive forces that lie in the Earth-Moon orbital plane? Are the spacecraft out-of-plane dynamics controllable using propulsive forces that are normal to the Earth-Moon orbital plane?
6. Design an orbit transfer maneuver that transfers the spacecraft from the liberation point L_2 to the state specified in Problem 4 with transfer time of 2.5 days.
 - a. Describe an orbital maneuver for the spacecraft that uses two propulsive force impulses, assuming the first impulse occurs when the maneuver is initiated and the second impulse occurs when the maneuver is terminated. For each impulse, describe the direction and magnitude of the instantaneous change in the spacecraft velocity vector that achieve the desired result. Verify through analysis or simulation that the orbital maneuver objective achieved using your propulsive force scheme.
 - b. Describe an orbital maneuver for the spacecraft that uses a non-impulsive propulsive force. Describe the direction and magnitude of propulsive forces that achieve the desired result. Verify through analysis or simulation that the orbital maneuver objective is achieved using your propulsive force scheme.